

My report Title

*Phase II project report submitted in partial fulfilment of the requirements for
the award of the degree of*

Bachelor of Technology

in

Computer Science & Design

Submitted by

My Name

Teammate1 Name

Teammate2 Name

Teammate3 Name



Focus on Excellence

Federal Institute of Science And Technology (FISAT)®
Angamaly, Ernakulam

Affiliated to

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CET Campus, Thiruvananthapuram
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FEDERAL INSTITUTE OF SCIENCE AND TECHNOLOGY
(FISAT)

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CERTIFICATE

This is to certify that the project phase II report for the project entitled “**My report Title**” is a bonafide report of the project presented during VIIIth semester (CXD416 - Project Phase II) by **My Name(FITCSD2001)**, in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology (B.Tech) in Computer Science & Design during the academic year 2025-26.

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Project Coordinator

Project Guide

Head of the Department

ABSTRACT

Abstract is a condensed version of your whole project.

- It should cover the major parts of the project.
- It should not contain any excess wordiness or unnecessary information.
- It should be readable and well organized.
- It flows smoothly between the parts.

An abstract convey the following information:

- The purpose of the project, identifying the area of study to which it belongs.
- The problem addressed.
- The research problem that motivates the project.
- Its importance/novelty
- The methods/approach used to address this problem, documents or evidence analyzed, highlighting novelty, if any.
- the major results obtained
- The conclusions reached or, if the work is in progress, what the preliminary results of the investigation suggest.
- The significance of the project. Why are the results useful? What is new to our understanding as the result of your inquiry?

Contribution by Author

Author1 Contribution

Author2 Contribution

Author3 Contribution

Author4 Contribution Goes Here

Student1 Name

Student2 Name

Student3 Name

Student4 Name

ACKNOWLEDGMENT

Your Acknowledgment Goes Here

In the acknowledgment section, you will recognize the support and guidance of the people who have helped you with any aspects of the project.

Student 1

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Chapter 1

Introduction

1.1 Overview

Give an overview of your project.

- what is the problem/question you are investigating?
- what is the starting point?
 - The evidence that there is a problem
 - Open questions in the research literature
 - new techniques/tools that need evaluation
- what have you done about it?
- Social relevance of your project.
- Applications
- Ethical and Professional Considerations

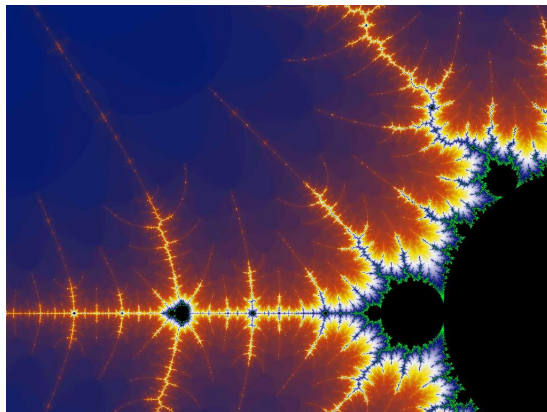


Figure 1.1: Mandelbrot Fractal

Overview can cover atleast one and half page.

1.2 Problem Statement

A problem statement is a concise statement describing the problem or issue addressed by the study. The problem should be composed in a way that both experts and non-experts in the field can understand.

A problem statement is a brief overview of the issues or problems existing in the concerned area selected for the research. It is an explanation of the issues prevalent in a particular sector that drives the researcher to take interest in that sector for in-depth study and analysis, so as to understand and solve them.

The word count of the problem statement should be in the range of 150-200 words. The problem statement in any work, includes four important segments,

- Background of the Problem: Here you can reflect on facts related to the problem to make the reader understand about the gravity of the problem.
- Anchor: How one needs to resolve this problem in the research paper.
- General problem: How it impacts a larger population.
- Specific problem: How it impacts your sample population

1.3 Objectives

Your project report will be bursting with essential information about your project. Although the content of your report will differ depending on the type of report you're creating, keeping your report organized will make it easy for the reader to follow along without missing any critical points.

1.4 Scope of the project

Give scope of the work.

1.5 Social Relevance

Explain social relevance of the project.

1.6 Organization of the report

Organize your data and content into sections that allow all stakeholders to quickly reference.

Chapter 2

Literature Review

After giving a brief explanation about the project and its significance, explain the related works.

2.1 Related work

Must include proper citations. asdas [?] and asdsad [?]

2.2 Comparison of related works

Give comparison charts/graphs/tables of the survey papers.

2.3 Proposed System

Explain proposed system

Chapter 3

Design

3.1 Functional Requirements

3.2 Non-Functional Requirements

3.3 Design Methodologies

3.4 System Architecture

3.5 Flow Chart

3.6 Use-Case Diagram/Data Flow Diagram

Chapter 4

Implementation & Testing

4.1 Implementation Details

4.1.1 Dataset

4.2 Libraries/Applications

4.3 Testing

All project modules must be unit tested. Testing should cover both backend and frontend components to ensure correctness and reliability of the system. You do not have write theory about what unit test and explain terminology.

4.3.1 Scope of Testing

Unit tests must be written for:

- Backend modules (APIs, business logic, database operations, utility functions)
- Frontend modules (UI components, client-side logic, API integration)
- Core algorithms or processing modules (if applicable)

Each logical function or component must have corresponding automated test cases.

4.3.2 Minimum Test Case Requirement

For each module or function, students must include at least the following test cases:

- Normal Case – Valid input producing expected output
- Boundary Case – Edge values or limits
- Negative Case – Invalid input or incorrect parameters
- Exception Case – Proper handling of errors or failures

- Invalid data type input
- Missing parameters
- Out-of-range values
- API failure responses

4.3.3 Test Execution

Tests must be automated and reproducible. All tests should run using a single command such as: `pytest` or `npm test`

4.3.4 Coverage Requirement

Students must generate a code coverage report using appropriate tools such as:

- `pytest-cov` (Python)
- Jest / Istanbul (JavaScript)

Chapter 5

Results

5.1 Sample

5.2 Comparison

5.3 Sustainability Impact Assessment

(Instructions for Students...)

This section should explain how the implemented system contributes to the selected Sustainable Development Goals (SDGs) based on the actual implementation and evaluation results obtained in Phase II.

Students should avoid repeating the theoretical discussion already presented in Phase I and instead focus on practical impact, measurable outcomes and future relevance.

5.3.1 Contribution of Project Features to SDGs

Under this subsection, student should explain how specific features or modules of the implemented system support the selected SDG(s).

You should include:

- A brief mention of the SDG(s) selected for the project.
- Description of key features/modules implemented in the system.
- Explanation of how each feature contributes to the SDG(s).
- Focus on practical implementation aspects rather than theory.
- Students may discuss points such as:
 - Automation that reduces manual effort or resource usage
 - Improved efficiency or productivity
 - Better accessibility of services to users
 - Support for entrepreneurship, employment or digital services
 - Resource sharing, reuse or reduction of waste

- Responsible use of data and technology

Students should clearly connect system functionality to sustainability outcomes.

5.3.2 Evaluation Results and SDG Impact

In this subsection, students should explain how the performance results obtained from testing and evaluation demonstrate the project's contribution to the selected SDGs.

Students should include:

- Key evaluation metrics or testing results (accuracy, efficiency, response time, utilization rate, etc.)
- Explanation of how these results support sustainability goals
- Evidence showing improvement in efficiency, accessibility, productivity or resource optimization
- Brief interpretation of results rather than repeating raw data.
 - Key evaluation metrics or testing results (accuracy, efficiency, response time, utilization rate, etc.)
 - Explanation of how these results support sustainability goals
 - Evidence showing improvement in efficiency, accessibility, productivity, or resource optimization
 - Brief interpretation of results rather than repeating raw data.
 - Students should focus on:
 - How the system improves performance compared to existing/manual methods
 - How the results support responsible resource usage or economic/social impact
 - Any measurable improvement achieved through the implemented solution.
 - Any measurable improvement achieved through the implemented solution.

5.3.3 Future Improvements and SDG Impact

This subsection should describe how the project can be further improved or scaled to enhance its contribution to the SDGs in the long term.

Students should include:

- Possible technical improvements or additional features
- Opportunities for scaling the system to larger users or real-world environments

- Enhancements that could increase efficiency, accessibility, or sustainability impact
- Integration with advanced technologies, larger datasets, or cloud platforms
- Potential societal, economic, or environmental benefits in the long term

Students should explain how these future developments would strengthen the project's contribution to sustainable development.

Chapter 6

Conclusion

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References

Appendices

Appendix A

CODE

Appendix B

Screenshots